

MKH-650模块化四轴搬运机械手项目书

20 kg负载 | 1180 mm工作半径

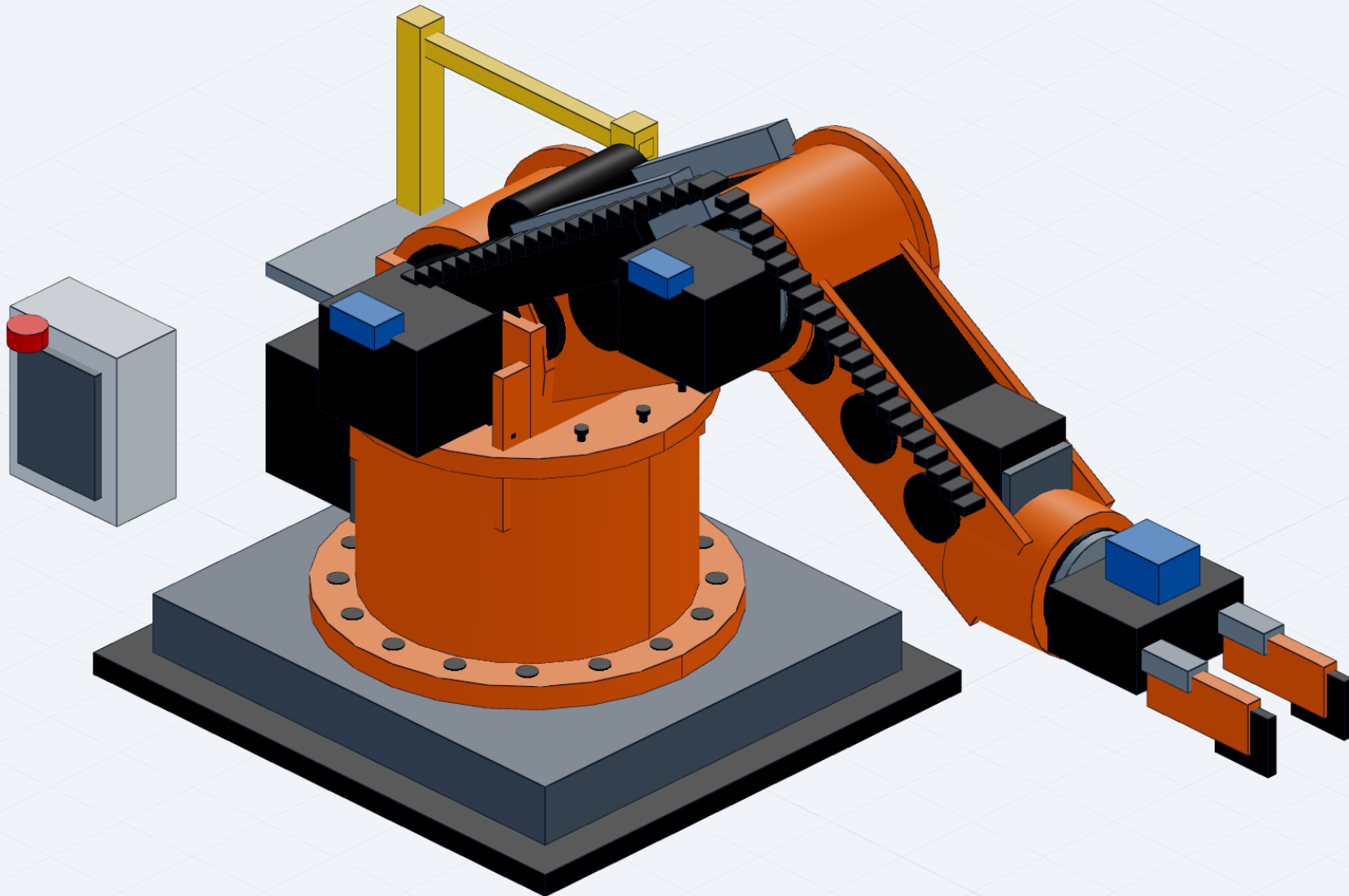
面向重载搬运的模块化四轴机械手，覆盖大型关节、箱式连杆、平衡机构、重载夹爪和维护设计。

16点基础锚固与J1回转支承
J2/J3复合载荷关节与12点法兰
七道筋板箱式连杆与平衡机构
1.8 kN重载平行夹爪设计目标

Editable CAD source + STEP + GLB + calculations + BOM + manufacturing plan + validation evidence

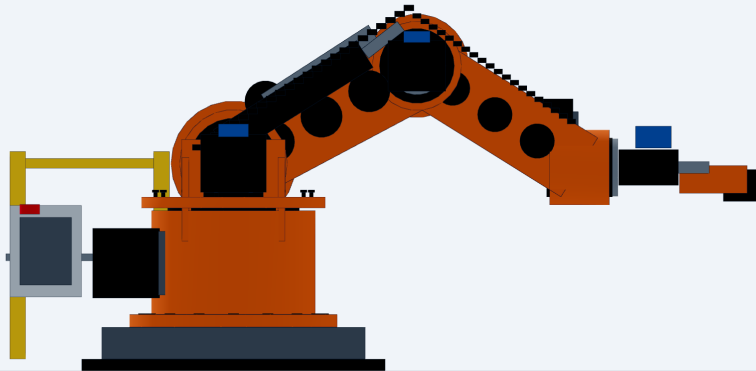
01 Detailed assembly overview

ISO

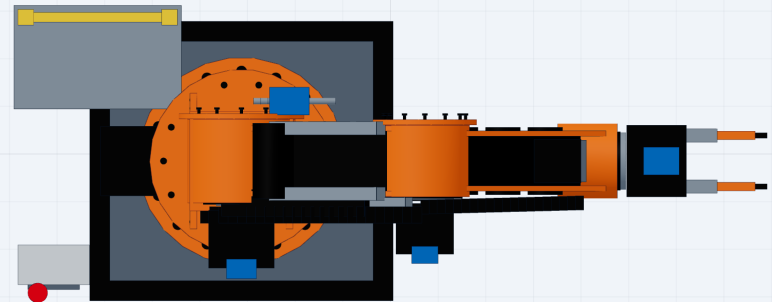


02 Orthographic and opposite views

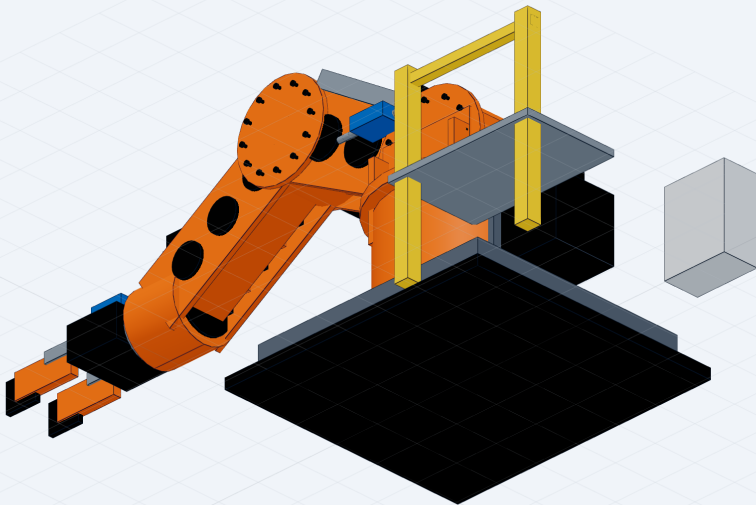
FRONT



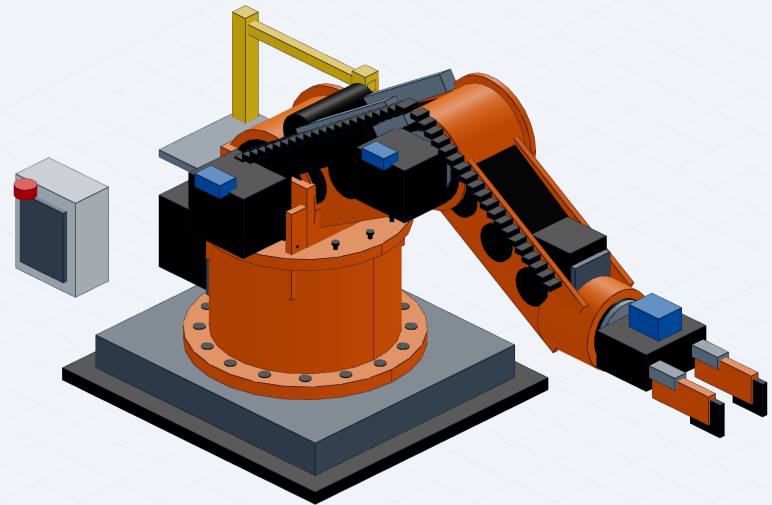
TOP



ISO



ISO



03 Parameters and preliminary calculations

parameter	value	unit	status
rated_payload	20	kg	report/resume baseline
upper_arm_length	520	mm	assumed dimension chain
forearm_length	480	mm	assumed dimension chain
tool_offset	180	mm	assumed dimension chain
maximum_radius	1180	mm	calculated
preliminary_J2_design_torque	1552	Nm	calculated
gripper_design_force	1770	N	calculated baseline
model_bounding_box	1886. 2x744. 8x916 . 6	mm	verified
step_faces	1047	count	verified
step_edges	1653	count	verified

$$R = 520 + 480 + 180 = 1180 \text{ mm.}$$

Estimated horizontal-extension static moment includes 20 kg payload, 18 kg tool, 24 kg forearm and 38 kg upper arm at their respective centers: approximately 693 N*m. With dynamic factor 1.6 and service factor 1.4, preliminary J2 output design torque is 1552 N*m.

For 20 kg payload, friction coefficient 0.25 and safety factor 2.25: $F = m g S / \mu = 1766 \text{ N}$. The gripper is therefore designed around 1.8 kN total clamping force, with replaceable serrated pads and fail-safe pressure monitoring.

The upper arm and forearm use twin 14 mm side plates, box spines, seven internal cross ribs and local flange reinforcement. Final plate thickness and rib geometry require FEA and weld/machining route review.

04 BOM, manufacturing and validation

item	component	qty	material_or_spec
1	foundation plate	1	Q235B
2	grouted base block	1	welded steel
3	J1 slew housing	1	QT500/steel fabrication
4	J1 slew bearing	1	cross roller/slewing bearing
5	shoulder tower	1	welded steel or cast iron
6	J2 joint module	1	servo + reducer + cross roller bearing
7	upper arm side plates	2	Q345B or 7075 aluminium option
8	upper arm ribs/covers	10	matching structure material
9	J3 joint module	1	servo + reducer + cross roller bearing
10	forearm side plates	2	Q345B or 7075 aluminium option
11	forearm ribs/covers	10	matching structure material
12	balance linkage	1 set	40Cr pins + steel links
13	gas spring/balancer	1	industrial rated
14	J4 wrist module	1	servo + planetary reducer
15	heavy parallel gripper	1	1.8 kN target
16	replaceable fingers	2	40Cr + serrated pads
17	robot cable chains	2	high-flex
18	central lubrication block	1	8-port
19	terminal box	1	IP54+
20	maintenance platform	1	steel
21	high-strength flange bolts	1 set	10.9/12.9 grade

Manufacturing / assembly

Large base and tower parts use welded or cast blanks followed by stress relief and finish machining of rotary datums. Joint housings are finish bored in one setup where possible. Pins and output shafts use 40Cr quenched and tempered steel with ground bearing seats. Arm plates are CNC profiled, ribbed and dowel located. Critical flanges use pilot diameters, dowel pins and high-strength bolts. Assembly proceeds from J1 upward with runout, backlash and

cable-bend checks at every joint.

Validation boundary